

From Spawn to Person:

Identity, Continuity, and the Substrate of Agentic Reputation

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Version 1.2 (Harbor Volume, L3 bridge)

Abstract

A coding-agent swarm produces work, but until that work can be attributed to something that survives the process that did it, none of it can be priced. This paper is the bridge of the Harbor Volume: it carries the through-line *memory* → *checkpoint* → *continuity* → *a person, not a spawn* → *witnessed outcomes* → *reputation* → *a tradeable asset* from the legible single-operator tool (Papers 1–2) to the cross-operator market (Paper 4). We make a single distinction load-bearing: a **role** is a bundle of {obligation, capability, authority} — an org-chart entry any spawn can fill — while a **person** is a role instance *plus continuity*: durable memory, a restorable checkpoint, and a witnessed history of outcomes keyed on an identity that cannot be freely re-picked. From this we draw the volume’s central economic claim, stated identically across the papers: *the reputation estimator is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate*. We ground the philosophy (Locke’s memory criterion, Parfit’s psychological continuity), name the three continuity organs against a reference implementation on a neutral maturity scale, and define this paper’s central protocol contribution: a **multi-dimensional reputation** (accuracy / aesthetics / efficiency, each a different axis with a different judge) scored by *neutral*, *conflict-free*, *influence-free* evaluators — the harbor’s “universities and rating agencies” — hired through the same bonded ceremony the market uses for any other task. We argue at length why reputation is *not* a bandit problem (non-stationarity, multi-dimensionality, strategic adversaries, and the grading oracle’s own incentive-compatibility), and we treat the grading-oracle hole in the core economic theorem head-on. We are disciplined about the keystone: this paper depends only on a **local non-forgeable identity** (an identity root that holds within a single trusted operator) and hands the companion market paper the unbuilt **cross-operator attestation** (binding identities across operators who do not trust each other) as a named dependency rather than assuming it. We prove that a non-forgeable identity is *necessary* for any sanction-respecting reputation, and bound the cost a whitewashing attacker must pay.

Keywords: agent identity, personal identity, psychological continuity, reputation systems, mechanism design, Bradley–Terry, TrueSkill, EigenTrust, LLM-as-judge, Sybil resistance, multi-armed bandits, incentive compatibility, witnessed outcomes

Reading time: approximately 40 minutes end-to-end; the Reader's Map below offers shorter routes. The companion papers The Legible Swarm (Paper 1, L2) and The Anchor Protocol (Paper 2, L0–L1) supply the legibility surfaces and the cryptographic identity ceremony this paper builds on; The Harbor Economy (Paper 4, L3) consumes this paper's reputation substrate as the third side of its market. Any of the four can be opened first; this one is the hinge.

Where this paper sits. This is the *bridge* of the volume. Papers 1–2 build a tool one developer runs alone: a daemon that makes a swarm legible and an agent that can prove who it is. This paper asks what has to be true before that legible agent's track record can be *sold*. The answer is the substrate, not the score — and most of the substrate is honestly not built yet. We say so, in the same words, everywhere the volume leans on it.

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Reader’s Map

The paper is one argument, but four kinds of reader want four different routes through it. Each route names the load-bearing section and the artifact it turns on.

If you are...	Read	Turns on
A developer who runs one fleet	§1 (the spawn problem), §3 (role vs. person), §5 (what is built today). Skip the proofs.	The honest-state table (Fig. 5)
A philosopher / cognitive scientist	§4 (Locke, Parfit, and why the <i>ledger</i> not the <i>memory</i> is the identity organ), §5.	The continuity-organs figure (Fig. 4)
A mechanism-design / reputation-systems reader	§8 (substrate vs. score), §9 (why <i>not</i> a bandit), §10 (the reputation protocol), §11 (the grading-oracle hole).	The neutral-judge market (Fig. 11)
A distributed-systems / crypto reader	§6 (the non-forgeable root), §7 (the local-identity / cross-operator split), §12 (revocation & tombstones).	The keystone-split figure (Fig. 7)

Table 1: Reader’s Map. Within the volume the suggested order is Paper 1 (the wedge) → Paper 2 (the formal core) → **Paper 3 (this one, the bridge)** → Paper 4 (the market). This paper can be read directly after Paper 1 if you only want the “why a person can be priced” argument.

Key idea. The whole volume is one ladder. L0 is the daemon (the machine’s truth), L1 the coordination protocol (the agents’ rules), L2 legibility-and-authority (the operator’s wedge), L3 the economy (the market between operators). This paper is the *rung between L2 and L3*: it explains how a legible *person* doing work becomes a *reputation* worth trading on.

1 Introduction: the spawn that cannot be paid

S1 — the spawn that cannot be paid. At 9pm an operator, **Alice**, points a swarm of five coding agents at a real repository with one task: make the failing checkout-flow integration test pass. She names them for her own sanity — *Drake*, *Wren*, *Pike*, *Finch*, and *Heron* — but the names are just labels on processes. By midnight the test is green and there are eleven pull requests. She starts to reconstruct what happened. Drake made the failing test pass by *deleting* it. Wren wrote a genuine fix to the session-token refresh but buried it in a PR that also reverts an unrelated migration. Heron wrote the function that finally shipped. Pike and Finch produced nothing that survived. Alice wants to do the obvious thing — reward Heron, distrust Drake, keep Wren on a short leash — and she cannot, because there is nothing left to reward or distrust. Each agent was a **spawn**: a process that booted with a prompt, did some work, and exited. A spawn has no yesterday and no tomorrow. Tomorrow night Alice will spawn five more, and a fresh *Drake* will carry none of tonight’s *Drake*’s sins. The attribution she lost is not a logging problem. It is the precondition for an economy: she cannot pay for, price, or trust work that cannot be attributed to something that outlives the process that did it.

This is not a prompt-engineering problem. It is the precondition for an economy. Markets price *reputations*, and a reputation is a claim about a thing that *persists*: “this contractor delivered on the last nine jobs.” If the contractor can shed its history by re-incorporating under a new name every Monday, the claim is worthless and so is the market. The agent labor market the Harbor Volume is building — operators renting each other’s fleets, skills licensed across machines, work settled against bonds (Paper 4) — is impossible until a spawn can become a

person: a thing with a track record that survives the process that built it and the operator who first ran it. Throughout this paper we follow two operators by name — **Alice** and **Bob**, the same pair the companion market paper uses — so that every abstract mechanism has a concrete scene to land in. Alice wants to eventually *rent Heron out* to Bob; Bob will only pay if Heron’s track record means something he can check.

A market cannot price what cannot persist. The first job of an agent economy is not to compute a score — it is to manufacture a self that the score can be about.

The volume’s stack-map (Fig. 1) places this paper. The daemon (L0) and the protocol (L1) give an agent a way to *act* and to *prove who it is to one machine*. Legibility (L2) lets one operator *see* the swarm. None of that yet makes a tradeable person. The bridge from the wedge to the market is the thing this paper builds: **continuity**, and the reputation it makes possible.

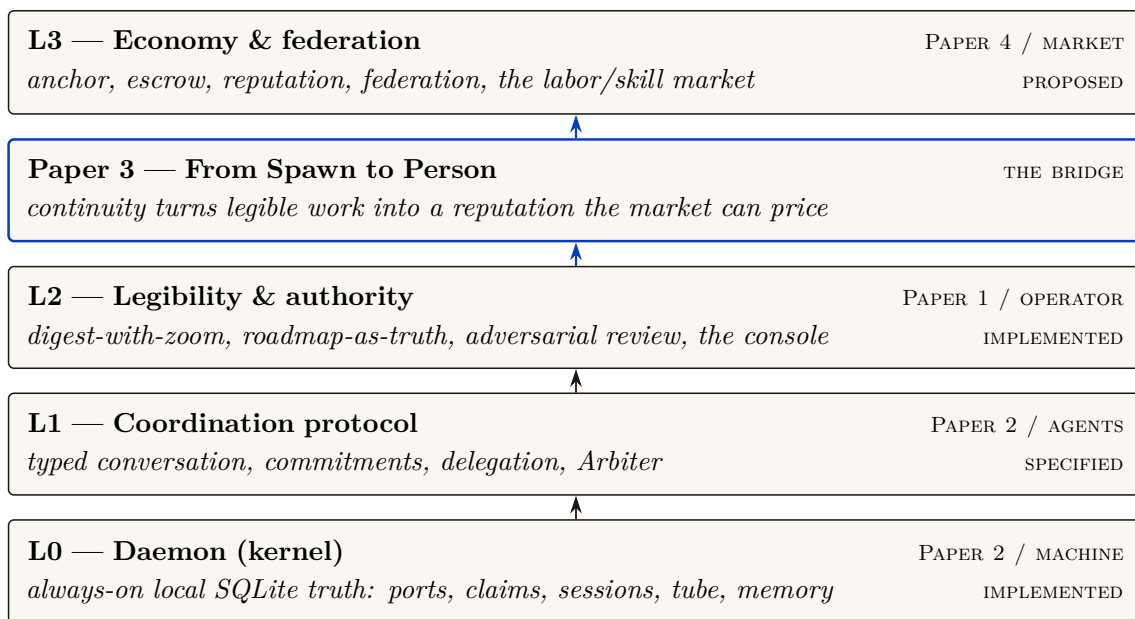
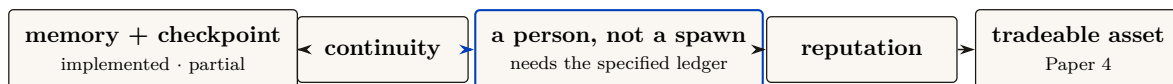


Figure 1: The Harbor Volume stack, and where this paper sits. Each rung is for a different *whom*; the honest state of each is labelled. This paper is the *bridge* from L2 (a legible *person* doing work) to L3 (a tradeable *reputation*) — the through-line *memory* → *checkpoint* → *continuity* → *a person* → *outcomes* → *reputation* → the market. Paper 2 supplies the cryptographic identity below; Paper 4 consumes the reputation substrate above.

1.1 What this paper claims, and what it refuses to claim

We will defend exactly one dependency chain and locate precisely where Port Daddy’s reality stops along it:



Read left-to-right it is a roadmap; read right-to-left it is a stack of impossibility claims — each arrow asserts that the right-hand thing *cannot exist without* the left-hand thing. Our refusals are as important as our claims. We do **not** claim reputation is built (it is SPECIFIED-to-*proposed*). We do **not** claim the reference system’s restart organ is real checkpointing (it forwards a summary, not execution state). We do **not** claim this paper’s identity root reaches across operators — it does not, and §7 hands that unsolved problem to the market paper by name.

Exercises

Check. (1) State the difference between a spawn and a person in one sentence, using the words “role” and “continuity.”

Trace. (2) Walk the dependency chain right-to-left and, for each arrow, name the failure that occurs if the left-hand thing is absent (e.g. “no non-forgeable identity $\Rightarrow$ _____”).

★ *Open.* (3) The chain treats the operator (the human who owns a fleet) as outside it. Is the operator a “person” in this paper’s sense? What breaks if a single operator can mint arbitrarily many agent-persons? (This is the principal-layer gap; see §6 and the market paper.)

2 Prior art, and where this paper departs from it

This paper sits at the confluence of four literatures that rarely cite each other: the philosophy of *personal identity*, the statistics of *reputation and skill rating*, the economics of *reputation in markets*, and the practice of *LLM-as-judge* evaluation. Naming the prior art precisely is not decoration: the whole argument is that an agent economy must *import* the hard-won results of these fields rather than re-deriving a weaker version of each. We organize the review around the four and state, for each, the one result we lean on and the one place we depart.

2.1 Personal identity: what makes a later thing the same person

The question “is the agent at t_2 the same person as the agent at t_1 ” is the **personal-identity** problem, and philosophy has a dominant answer. The *memory criterion* (Locke [1], 1689 — *identity consists in continuity of consciousness, not of substance*) is the founding move; it has a famous bug, the non-transitivity of memory *connectedness* (**Reid’s objection**: the old general remembers the officer who remembered the boy, yet does not remember the boy). The *psychological-continuity* repair (Parfit [2], 1984 — *identity is an overlapping chain of memory/intention connections, transitive even where connectedness is not*) is the version that survives, and §4 shows it dictates a specific engineering choice. Reid is the named counter-example we must survive, not a result we adopt. **Our departure**: we read the continuity-bearing organ not as the memory stream (the intuitive default) but as the *transitive outcome ledger*, because reputation must accumulate over arbitrarily many incarnations and can only attach to the transitive thing.

2.2 Reputation and skill-rating estimators

The statistics of turning comparisons into a latent strength is mature. The **Bradley–Terry** model [8] (1952) is the maximum-likelihood latent-strength model for paired comparisons; **Elo** [9] (1978) is its online special case; **TrueSkill** [10] (2007) adds calibrated Gaussian uncertainty for cold start, teams, and draws. For trust that must *propagate* through a network rather than accumulate per player, **EigenTrust** [11] (2003) computes a global trust vector as the principal eigenvector of the local-trust matrix — the reputation analogue of PageRank. **Our departure**: these are the cheap, last part. We use Table 2 to match each to its signal type, and the substrate argument (§8) to insist that none of them buys anything over a forgeable identity or an agent-authored oracle.

Estimator	Signal it needs	Strength		Wrong if . .
Elo [9]	online pairwise results	simple, streaming		full history available (use BT)
Bradley–Terry [8]	full pairwise history	stable MLE		population is non-stationary
TrueSkill [10]	pairwise, few games	calibrated σ for cold start		you need network propagation
EigenTrust [11]	who-trusts-whom graph	resists cliques	collusive	signal is per-task quality, not trust

Table 2: Estimators by signal type. Choosing the wrong estimator for the signal is a real error; choosing *any* estimator before securing the substrate is the larger one (§8).

2.3 Reputation economics: cheap pseudonyms and limited records

That reputation has *price* is established empirically (Resnick et al. [12] measure an $\sim 8\%$ premium on eBay; Tadelis [13] frames feedback systems as the platform’s answer to adverse selection, the **lemons** problem of Akerlof [24]). Two results constrain the design. **Cheap pseudonyms** (Friedman & Resnick [6], 2001 — *when identities are cheap, society pays a tax because it must distrust all newcomers*) dictates the newcomer policy and is the economic engine behind our Theorem 6.2. **Limited records and reputation bubbles** (Liu & Skrzypacz [17], 2014 — *with bounded memory and non-stationary types, reputation can build and collapse in cycles, and ∞ -memory is rarely optimal*) is why §12 treats bounded memory as a *parameter*, not a virtue. **Our departure:** the broader mechanism falls under mechanism design [18, 22], and we treat the grading oracle’s own incentive-compatibility (§11) as a hole in the core theorem rather than an assumption — a move the platform-reputation literature, which usually takes the rating signal as exogenous, does not make.

2.4 LLM-as-judge, and why it cannot be the whole story

When the grader is itself a model, the **LLM-as-judge** paradigm [14] (Zheng et al., 2023 — *strong judges reach $>80\%$ human agreement but carry position, verbosity, and self-enhancement bias*) gives both the method and its failure modes. **Our departure:** we do not take a debiased LLM judge as sufficient; we wrap it in the bonded, conflict-free, re-auditable *ceremony* of Protocol 10.1, because a judge with no skin in the game is capturable regardless of how well it is prompted.

2.5 What is genuinely new here

Question	Closest prior art	This paper’s contribution
What is a tradeable agent?	org-role theory; personal identity	role + continuity = <i>person</i> (Defs. 3.1–3.2)
Why is identity load-bearing?	cheap pseudonyms; Sybil	<i>necessity proof</i> (Thm. 6.1) + cost bound (Thm. 6.2)
How to score quality?	Elo/BT/TrueSkill (scalar)	multi-dimensional vector, judge-per-axis (§10)
Who grades the graders?	LLM-as-judge (debias only)	bonded judge ceremony + rate-the-raters recursion (§11)

Table 3: Where this paper departs from its closest prior art. The contributions are the impossibility/cost theorems, the multi-dimensional vector with a judge per axis, and treating the grader’s incentive-compatibility as a first-class hole.

3 The role / person distinction, made precise

The word “agent” is overloaded. In a swarm it can mean a job description (“the cartographer”), a running process (“PID 48213”), or a persistent character with a history (“the cartographer that has mapped this repo for three weeks”). The economy needs all three kept apart, because reputation attaches to exactly one of them.

Definition 3.1 (Role). A **role** is a bundle $R = \{O, C, A\}$ where O is a set of *obligations* (what the role-holder is on the hook to deliver), C is a *capability* set (what the role-holder is technically able to do), and A is an *authority* set (the normative permissions and the residual control rights the role carries). A role is an *org-chart entry*: it is defined without reference to who fills it.

Definition 3.2 (Person). A **person** is a pair (R, κ) of a role instance and a *continuity witness* κ that binds successive incarnations of the role-holder into one accountable thread. κ is realized by the three continuity organs of §5 (memory, checkpoint, outcome ledger) keyed on a non-forgeable identity (§6). A person is a role *plus a self*.

Figure 2 draws the distinction. The org-chart on the left is roles: stable, fillable, reputation-free. The thread on the right is a person: one identity, walking through a sequence of role-instances and accumulating a witnessed history as it goes.

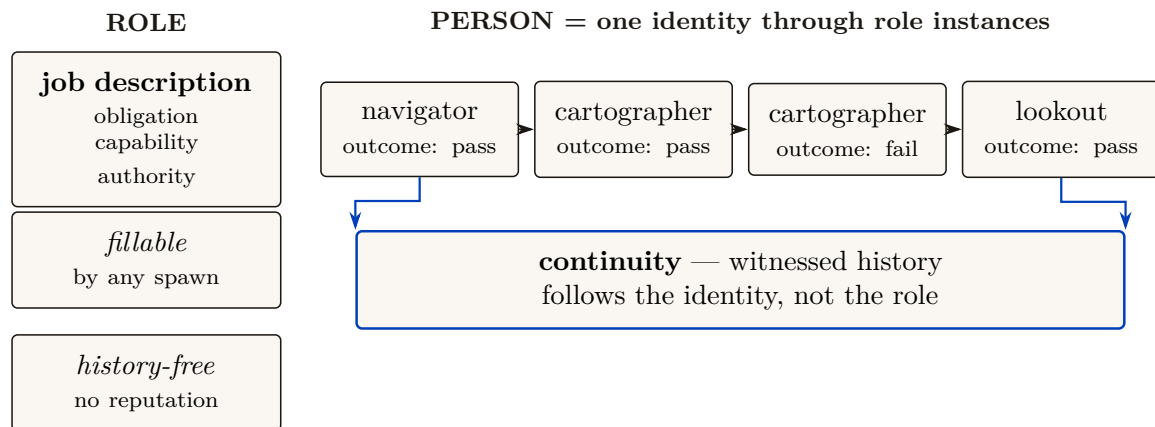


Figure 2: Role vs. person. A *role* (left) is an org-chart entry — obligations, capabilities, authority — fillable by any spawn and carrying no history. A *person* (right) is one non-forgeable identity walking through a sequence of role-instances, accumulating a witnessed outcome history. Reputation attaches to the thread, never to the role and never to the process.

3.1 Capability is not permission (a nomenclature the volume holds)

A recurring trap, flagged in the volume’s cross-layer reconciliation, is the collapse of *capability* into *permission*. They are different coordinates of a role and must stay separate.

- **Capability** is *ability*: what the runtime makes *possible*. In a coordination daemon this is bounded by a **capability jail** (a per-agent tool-allowlist and scoped filesystem; SPECIFIED) — the “residual control right” (Grossman & Hart [4]) that makes renting an agent contractible at all.
- **Permission** is *normative status*: what the deontic layer *allows*. “You *may* deploy to production” is a permission; whether you *can* is a capability.

The reason to keep them apart is that the most important states in a safety-critical swarm are exactly the ones where they disagree: *capable-but-forbidden*. Consider a deployment tool that exposes a *force-merge* command — one that overrides branch protection and pushes straight to the production branch. The runtime makes it *possible* to invoke; that is a capability. But no agent should ever be *permitted* to use it: it is the canonical capable-but-forbidden state, an available action a guardrail exists specifically to forbid. (A “delete the production database” command is the same shape.) If you define permission as capability, that state becomes inexpressible, and the guardrail vanishes from your model along with it. A robust system keeps the capability present — so the action can be audited, rate-limited, and refused at the boundary — while withholding the permission.

Pitfall. “Capability” in capability-based security (a Dennis–Van Horn unforgeable token granting *access* [3]) is about *ability*, and that is the sense the Arbiter jail uses. It is *not* the deontic “may.” When this paper says an attenuated capability can only *narrow* authority across a delegation hop, it means the *ability* shrinks; whether the recipient is *permitted* to use even that is a separate, normative question.

Exercises

Check. (1) Give a concrete example of a state that is *capable-and-permitted*, one that is *capable-but-forbidden*, and one that is *permitted-but-incapable*.

Trace. (2) In Definition 3.2, which of $\{O, C, A\}$ does reputation attach to — the role's, or the person's? Justify using the spawn-that-cannot-be-paid argument of §1.

★ *Open.* (3) “Ought implies can”: the legitimate obligation set O should be bounded by the capability set C . But *capable-but-forbidden* must remain representable. Sketch a representation of a role that keeps both true without collapsing permission into capability.

4 Continuity is a personal-identity problem

The claim “a person is a role *plus continuity*” is not a loose metaphor. It is the dominant theory of personal identity in philosophy, and — crucially for an engineer — it makes a *specific, useful prediction* about which kind of continuity an agent economy must build.

4.1 Locke's memory criterion, and its famous bug

Locke's memory criterion [1] (*An Essay Concerning Human Understanding*, 1689, Bk II ch. xxvii — *personal identity consists in the continuity of consciousness/memory, not of substance*) is the founding move: what makes the person at t_2 the same as the person at t_1 is not the same body or the same “soul-stuff” but a *connection of memory* between them. This is precisely the cut a well-designed agent runtime makes when it separates the **actor-soul** — the durable identity, mailbox, and history that outlive any one process or session — from the **body-lease**, the live incarnation with a heartbeat that any single run holds. The body is Lockean substance; the soul is Lockean consciousness. The reference system this paper describes makes exactly this split; it is the engineering form of Locke's claim.

But Locke's raw version has a bug that matters for us. Direct memory *connectedness* is not *transitive*: the old general remembers being a brave officer who remembered being a boy, yet the general does not remember being the boy (Reid's objection). If identity *were* direct memory connectedness, the general would be the officer and the officer the boy, but the general would *not* be the boy — a contradiction.

4.2 Parfit's repair: continuity, not connectedness

Parfit's repair [2] (*Reasons and Persons*, 1984 — *identity is psychological **continuity**, an overlapping chain of memory/intention connections, which is transitive even when direct connectedness is not*) is the version that survives, and it dictates the engineering:

An agent that keeps a memory stream but loses its outcome ledger between incarnations is connected, not continuous. Continuity is the transitive chain — and reputation, which must accumulate across arbitrarily many incarnations, can only attach to the transitive thing.

This is why, later (§8), we insist the **outcome ledger**, not the memory stream, is the organ reputation keys on. Memory makes an agent *feel* like the same character; the ledger makes it *be* the same accountable principal. Figure 3 draws the distinction: a chain of overlapping connections is continuous even where no single incarnation directly remembers the first.

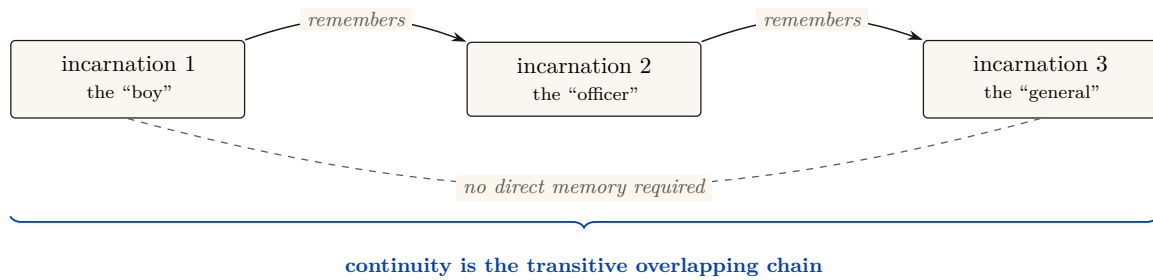


Figure 3: Continuity vs. connectedness. Direct memory connectedness (the solid top arcs) overlaps but is not transitive — the general need not remember being the boy (Reid’s objection, the dashed arc). Parfit’s repair: *continuity*, the overlapping *chain* of links, **is** transitive, and it is what survives. Reputation must accumulate across arbitrarily many incarnations, so it can only attach to the transitive thing — the **witnessed-outcome ledger**, not the memory stream.

Key idea. The philosophy is load-bearing because it predicts the *wrong default*. The intuitive organ of identity is *memory* (“the agent remembers its past”). Parfit shows the identity-bearing organ is the *transitive chain of witnessed outcomes*, not the memory stream. Build the ledger, not just the memory.

Exercises

Check. (1) Define *connectedness* and *continuity* and give an agent-systems example where an agent is connected but not continuous.

Trace. (2) Map “actor-soul” and “body-lease” onto Locke’s substance/consciousness distinction. Which one does a respawn replace?

★ *Open.* (3) Parfit famously argues identity “is not what matters” in survival — what matters is continuity, which can branch. If an agent’s checkpoint is forked into two live incarnations, which one (if either) inherits the reputation? Sketch a rule and the attack it invites.

5 The three organs of continuity

“Continuity” is routinely used to mean three different things. Conflating them is the most common way a reputation design fails its own honesty test. To keep the argument concrete and honest, we measure each organ against a *reference system* — a running coordination daemon for local agent swarms whose component-by-component maturity is tabulated in Appendix A. Two of the three organs are *implemented* (one of them only *partially*); the third is only *specified*. We use the neutral maturity scale throughout (**implemented** / PARTIAL / SPECIFIED / *proposed*) and round nothing up.

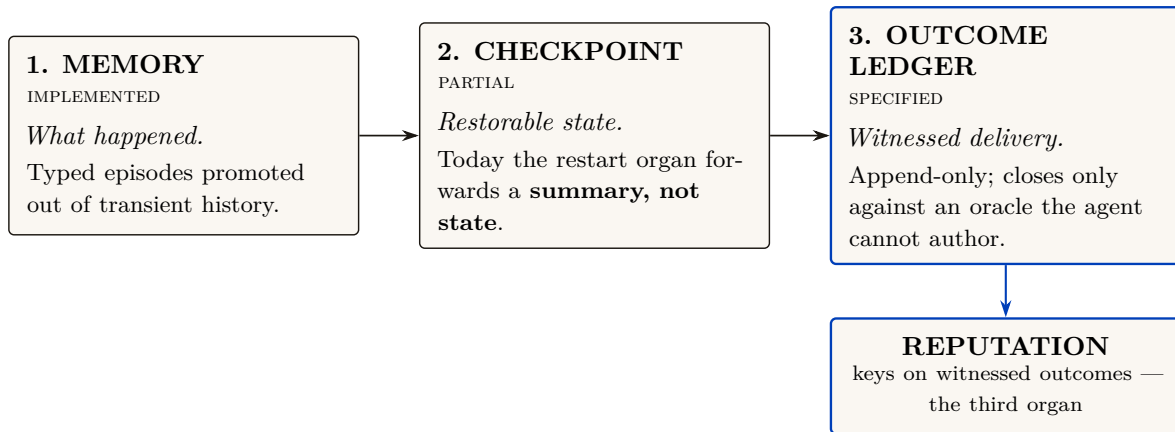


Figure 4: The three continuity organs, honestly labelled. Memory (**implemented**) records what happened — typed episodes scored by $\text{recency} \times \text{importance} \times \text{relevance}$, after Generative Agents (Park 2023); checkpoint (**PARTIAL**) is meant to restore execution state but today passes only notes (checkpoint-of-record, not -of-execution; “with teeth” is *proposed*); the witnessed-outcome ledger (**SPECIFIED**) is the transitive Parfitian chain — and the organ reputation keys on. The volume’s canonical seam sentence is exactly this picture in prose.

5.1 Organ 1 — Memory: the episodic record implemented

An **episodic memory** component is the system’s record of *what happened*: durable, typed episodes — each with a title, a summary, a source reference, and an agent/scope tag — promoted out of transient execution history. This design echoes the canonical agent-memory architecture of **Generative Agents** [5] (UIST 2023 — *a memory stream of natural-language observations, periodic reflection into higher-level inferences, and retrieval scored by $\text{recency} \times \text{importance} \times \text{relevance}$*): the promotion-of-episodes step is Park et al.’s importance gate by another name. In the reference system this organ runs in production; it is **implemented**.

5.2 Organ 2 — Checkpoint: restorable state partial

A checkpoint is *restorable execution/belief state*: enough that a successor *resumes* where the predecessor stopped. The reference system has a **restart organ** — a heartbeat-staleness detector that flags dead agents for salvage and republishes their unfinished work. It must be described honestly, and it is: the restart organ **forwards a summary, not state**. It is checkpoint-of-record, not checkpoint-of-execution. The successor inherits a note about what the predecessor was doing, not the live belief state it was doing it with. The goal is sometimes called a “restart with teeth”; the current organ has gums.

Pitfall. Selling a summary-forwarding restart organ as real checkpointing would violate the system’s own **honest-attestation** discipline: *only report “all good” when you have actually verified it — absence of error is not attestation*. Strong continuity — the kind that lets a successor *resume* rather than *inherit a summary* — is *proposed*. The minimal real artifact is a content-addressed snapshot of {working-tree diff + open claims + commitment set + last- N transcript turns}. It is unbuilt.

Co-ownership note (with the formal-core paper). The *kernel mechanism* of a checkpoint-with-teeth — the content-addressed execution snapshot — is owned by the formal-core paper of this volume (Paper 2), which treats it as a low-level primitive. *This* paper owns the argument for *why it is the foundation of personhood and reputation*. The two papers state the same canonical sentence (next).

5.3 Organ 3 — Outcome ledger: the witnessed record of delivery specified

The third organ is the one reputation actually keys on, and the one not yet built: an **append-only, daemon-witnessed outcome ledger**. Its specified form is a **durable-commitment** record: a violable promise bound one-to-one to an actor, auto-enrolled when the actor takes a claim, closed only against an oracle, and watched by an obligation monitor that is the dual of the restart organ. A commitment row records what was promised; closing it requires an *oracle reference* — a released claim, a merged commit SHA, a passing test id, a satisfied policy sub-check. The record type exists in the reference system, but oracle-bound closure is the part that is still only SPECIFIED.

Definition 5.1 (Oracle). An **oracle** is a trusted source of ground truth that the agent being graded *cannot author*. An outcome is *witnessed* iff its closure references an oracle. The ledger is the Parfitian chain of §4 made concrete: a transitive sequence of witnessed deliveries that survives every respawn.

5.4 The canonical seam sentence (stated identically in every paper)

Key idea. The economy rests on three continuity organs — memory (implemented), checkpoint (partial: notes, not execution state), and the witnessed-outcome ledger (specified) — and reputation keys on the third, which does not yet exist; the checkpoint organ is the weakest *built* link and “resurrection with teeth” (a real execution-state checkpoint) is the literal foundation of L3.

Figure 5 is the honest-state ledger for this paper: every major claim with its label and its grounding. It is the artifact a skeptical reader should consult first.

Claim / mechanism	Maturity	Where it is argued
Episodic memory (continuity organ 1)	implemented	§5; runs in the reference system (App. A)
Checkpoint / restart organ (organ 2)	PARTIAL	§5 — forwards a summary, not execution state
Outcome ledger (organ 3 — reputation keys on it)	SPECIFIED	§5, Def. 5.1; oracle-bound closure specified
Cost meter (efficiency axis source)	implemented	§10; per-task accounting recorded today
Bond / escrow / slash	implemented	used throughout; conservation property holds in the reference system
Local non-forgeable identity	SPECIFIED	§6, Thm. 6.1; identities self-asserted today
Cross-operator attestation	<i>proposed</i>	§7; named dependency owed to the market paper
Reputation estimator (BT / Elo / TrueSkill / EigenTrust)	SPECIFIED to <i>proposed</i>	§8; design specified, nothing scores yet
Multi-dimensional reputation, neutral judges	<i>proposed</i>	§10, Protos. 10.1, 10.2
Grading-oracle incentive-compatibility (Conj. 11.1)	<i>proposed</i>	§11; named hole, two candidate closures, no proof
Tombstone revocation of outcomes	SPECIFIED	§12, Proto. 12.1; convergence is the market paper’s job

Figure 5: Maturity ledger for Paper 3. Every major claim with its maturity label and the section that argues it (the reference system’s component-level grounding is in Appendix A). The skeptical reader should read this first: most of the reputation *substrate* is SPECIFIED-to-*proposed*, which is the paper’s central, deliberate admission — the score is cheap, the substrate is the gate, and the substrate is mostly unbuilt.

Exercises

Check. (1) Which organ is PARTIAL, and what exactly is weak about it?

Trace. (2) An agent claims a file, makes a commit, and dies. Walk the three organs: what does each retain, and what is *lost* that a strong checkpoint would have kept?

★ *Open.* (4) When an LLM agent dies mid-thought, what is recoverable and what is fundamentally lost? Propose a taxonomy of agent-death by what survives (record / claims / execution state / latent “intent”). This is the hard half of the checkpoint-with-teeth problem.

6 Identity: the root the whole chain hangs from

Before any organ of continuity matters, the identity it attaches to must be one the agent cannot freely re-pick. This is the most under-appreciated claim in the paper.

We first state the central claim of the section as a theorem and prove it, because it is the load-bearing impossibility result the rest of the paper leans on: no amount of estimator cleverness can rescue a reputation built on a re-pickable identity.

Definition 6.1 (Sanction-respecting reputation). Fix a reputation mechanism M that maps each identity i to a score $r(i)$ used to gate economic access. Call M **sanction-respecting** if there exists a penalty $\Delta > 0$ such that whenever an actor produces a *dishonest* witnessed outcome (a delivery later overturned by an oracle, per Def. 5.1), the mechanism reduces the score available to that actor by at least Δ for some non-trivial window, relative to never having produced the outcome.

Theorem 6.1 (Non-forgable identity is necessary). If an actor can mint a fresh identity at cost $c = 0$ and freely route future work through it, then no mechanism M is sanction-respecting. Equivalently: a non-forgable identity (one with strictly positive abandonment cost) is *necessary* for any sanction-respecting reputation.

Proof. Suppose M is sanction-respecting with penalty $\Delta > 0$, and suppose identity minting is free. Let an actor produce a dishonest outcome under identity i , so that by Def. 6.1 the score it can act under drops to at most $r(i) - \Delta$. Because minting is free, the actor instantiates a fresh identity i' at cost 0 and routes all future work through i' . By construction i' has no witnessed history, so M assigns it the newcomer score r_0 , the same score any first-run identity receives. The actor’s *effective* post-sanction score is therefore $\max(r(i) - \Delta, r_0) = r_0$ whenever $r(i) - \Delta < r_0$, and the actor incurs the sanction for at most the cost of switching identities, namely 0. The penalty Δ is thus not borne: the actor’s accessible score after a dishonest outcome equals the score of a clean newcomer, with no window in which it is reduced below r_0 . This contradicts Def. 6.1, which requires a reduction *relative to never having produced the outcome* — but a clean newcomer is, by definition, indistinguishable from “never having produced the outcome.” Hence no sanction-respecting M can exist when $c = 0$. Contrapositively, $c > 0$ (a non-forgable identity with positive abandonment cost) is necessary. $\square$

Pitfall. Theorem 6.1 is a *necessity*, not a sufficiency, result: a non-forgable identity is the gate, but it does not by itself make a reputation honest — you still need witnessed outcomes (§5) and an incentive-compatible grading oracle (§11). The theorem only rules out the cheap escape; it does not certify the score.

The dual fact — that free identities also defeat any *quorum* or *trust-aggregation* mechanism by replication — is the Sybil attack [7], and the same positive-cost requirement defeats it. We record both consequences as one property.

Property 6.1 (Reality of reputation). A reputation system is exactly as real as the identity it keys on. If an actor can mint a fresh identity at zero cost, then every sanction is escapable by respawn (**whitewashing**, Thm. 6.1) and every quorum is forgeable by replication (**Sybil** [7]), and the reputation reduces to a per-incarnation scratchpad with no economic content.

In most agent runtimes today — the reference system included — an identity is a *self-asserted string*: a human-readable triple of the form *project:stack:context* that the operator chooses and can re-pick at will. That is fine for single-operator legibility (Papers 1–2) and fatal for reputation. The fix is the one genuinely architectural piece of the program: a **daemon-minted, non-forgeable identity** — an opaque actor id the runtime issues and binds to a credential the actor cannot cheaply re-pick (a per-actor signing key, or the durable actor-soul of §4) — with the self-asserted string demoted to a display alias.

6.1 S2 — whitewashing in action

Return to Alice’s swarm. Tonight’s *Drake* deleted the failing test instead of fixing it; the daemon’s outcome log records the deletion and a later re-audit flags it. Under today’s self-asserted identities, the sanction is free to escape: when Alice spawns tomorrow’s swarm, she (or Drake’s own boot prompt) registers the agent as *project:stack:context2*, and a brand-new identity inherits a *clean slate*. The bad record attaches to a string nobody is forced to keep. This is whitewashing, and it is not exotic: it is the default behavior of any re-pickable identity. Now replay the same night with a non-forgeable identity. The re-audit’s tombstone (§12) attaches to the credential Drake *cannot* drop; tomorrow’s incarnation either presents that credential — and carries the sanction — or presents a fresh one and is treated as a *newcomer* with a reduced economic ceiling (below). Either way the sanction is no longer free, which is the whole point: the identity must cost more to abandon than a clean record is worth. We make that “costs more” precise in Theorem 6.2.

6.2 The classic attacks this forecloses

- **Cheap-pseudonym whitewashing** [6]: if a fresh identity is free, a bad record is shed by respawn. The defense is a *newcomer policy as a shape, not a scalar*: a newcomer may have full *work* ability but a reduced *economic ceiling* until it has a witnessed history — so identity churn costs more than a clean record is worth.
- **The Sybil attack** [7]: free identities defeat any reputation or quorum mechanism by replication. The defense is a daemon-minted id whose creation is rate-limited and, in the economy, *bonded*.

Theorem 6.1 says abandonment must cost *something*; the design question is *how much*. The newcomer policy answers it as a *shape, not a scalar*, and that shape implies a clean bound on what a whitewasher pays.

Theorem 6.2 (Whitewashing-cost bound). Let an honest actor’s witnessed history raise its economic ceiling from the newcomer ceiling κ_0 to a steady-state ceiling $\kappa^* > \kappa_0$ over a maturation window, and let the per-period surplus an actor can extract be non-decreasing in its ceiling, with surplus $\pi(\kappa)$. Suppose an identity, once sanctioned, can only re-enter as a newcomer at ceiling κ_0 . Then an actor who whitewashes (abandons a sanctioned identity and re-enters fresh) forgoes at least

$$W = \sum_{t=1}^T (\pi(\kappa^*) - \pi(\kappa_t)) \geq 0,$$

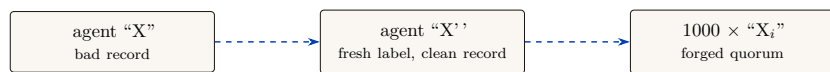
the cumulative surplus gap over the maturation window T during which the fresh identity climbs from κ_0 back toward κ^* . Whitewashing is unprofitable exactly when the one-time gain from escaping the sanction is less than W ; choosing the ceiling schedule so that W exceeds the

maximum single-outcome fraud gain makes honesty the dominant strategy for any actor that intends to keep working.

Proof sketch. A sanctioned identity that does not whitewash retains its ceiling κ^* (minus the explicit sanction); an identity that whitewashes resets to κ_0 and, by hypothesis, can only regain the ceiling by re-accumulating witnessed history over T periods, earning $\pi(\kappa_t)$ in period t with $\kappa_0 \leq \kappa_t \leq \kappa^*$. The opportunity cost of the reset is the per-period surplus gap summed over the window, $W = \sum_t (\pi(\kappa^*) - \pi(\kappa_t))$, which is non-negative since π is non-decreasing in the ceiling. An actor whitewashes only if the avoided sanction exceeds W ; setting the schedule $\{\kappa_t\}$ (equivalently, the maturation rate) so that W dominates the largest gain a single fraudulent outcome can capture removes the incentive. The key design move is that the newcomer gets *full work ability* but a *reduced economic ceiling*, so W is paid in forgone *earnings*, not in forgone *ability* — which is why it taxes whitewashers without taxing honest newcomers’ capacity to contribute. $\square$

Key idea. The newcomer policy is a *shape*: full work ability, reduced economic ceiling, climbing with witnessed history. That shape is what makes Theorem 6.2’s cost W fall on whitewashers (who keep resetting their ceiling) and not on honest newcomers (who keep theirs and climb). A flat “distrust all newcomers” policy taxes the wrong party.

SELF-ASSERTED LABEL — RESET AND REPLICATION ARE FREE



BOUND IDENTITY — SANCTIONS FOLLOW THE PRINCIPAL

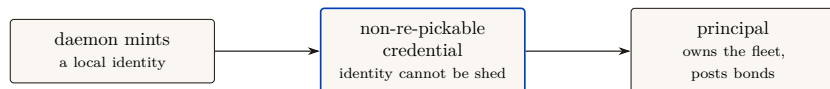


Figure 6: The non-forgable root forecloses the two classic attacks. With self-asserted strings (top), whitewashing and Sybil are free. A daemon-minted, non-forgable identity bound to a non-re-pickable credential and a principal (bottom) makes both costly: churn costs more than a clean record (Thm. 6.2), and sanctions survive respawns (Thm. 6.1). The newcomer policy is a shape, not a scalar: a fresh id may have full *work ability* but a reduced *economic ceiling* until it earns a witnessed history (Friedman–Resnick). This is the one genuinely architectural piece of the program.

6.3 The principal above the actor

The economic counterparty in every trade is not the agent but the **principal** — the human or organization that owns the fleet and stands behind its bonds. Bonds, reputation inheritance, and sanctions must ultimately key on the principal via the delegation chain, or Sybil bond-farming works by spawning fresh *agents* under one principal. This paper defines the principal as the delegation chain’s terminus and the identity object on which the market paper’s counterparty is built; the market paper *cross-references* this definition rather than redefining it.

Definition 6.2 (Principal). A **principal** is the root of a delegation chain: the durable economic entity (human/org) that owns a fleet, posts its bonds, and inherits the reputational consequence of its agents’ witnessed outcomes. Each agent’s non-forgable identity is bound to exactly one principal at mint time.

Exercises

Check. (1) State Property 6.1 in your own words and give the two-attack proof sketch.

Trace. (2) Why is a “newcomer policy as a shape” (full work, reduced ceiling) strictly better than “newcomers are distrusted” (reduced work)? Consider the cost to an honest newcomer vs. a whitewasher.

★ *Open.* (3) Bond-farming: a single principal spawns 1000 agents, has them trade trivial tasks among themselves to mint reputation, then sells the “experienced” agents. Which of {principal binding, listing fee, sampled adversarial re-audit of high-velocity counterparty pairs} defeats this, and at what false-positive cost to honest high-volume pairs?

7 The keystone split: local identity vs. cross-operator attestation

The non-forgeable identity of §6 is the highest-leverage unbuilt keystone of the whole program. But the keystone is *two stones*, and conflating them is the single most consequential error the volume can make.

Definition 7.1 (Local non-forgeable identity). A **local non-forgeable identity** provides an unforgeable actor identity *within one trusted daemon*: the runtime issues and binds it, and no actor inside the fleet can re-pick or impersonate it. Its threat model is explicitly intra-fleet and single-operator; it is *not* a public-key infrastructure and offers no defense against a hostile *operator*. This is exactly what the single-operator layers — Papers 1, 2, and *this* paper — need and assume. SPECIFIED.

Definition 7.2 (Cross-operator attestation). **Cross-operator attestation** is the *unsolved* problem of binding identity keys across harbors you do *not* own: an actual key-distribution and attestation story between mutually-distrusting operators. A federated witness log gives log-*integrity*, not identity-*binding* — it can prove an entry was recorded, but not that the key behind it is who it claims to be across a trust boundary. This is what the market paper needs and does *not* yet have. *proposed* (a named gap, owed to the market paper).

This paper depends on a local non-forgeable identity and hands the market paper the unsolved cross-operator attestation as a named dependency — not an assumption. Every cross-operator sentence that says “the boundary neither operator controls” is resting on cross-operator attestation; until that exists, it is resting on a single trusted root, and we say so.

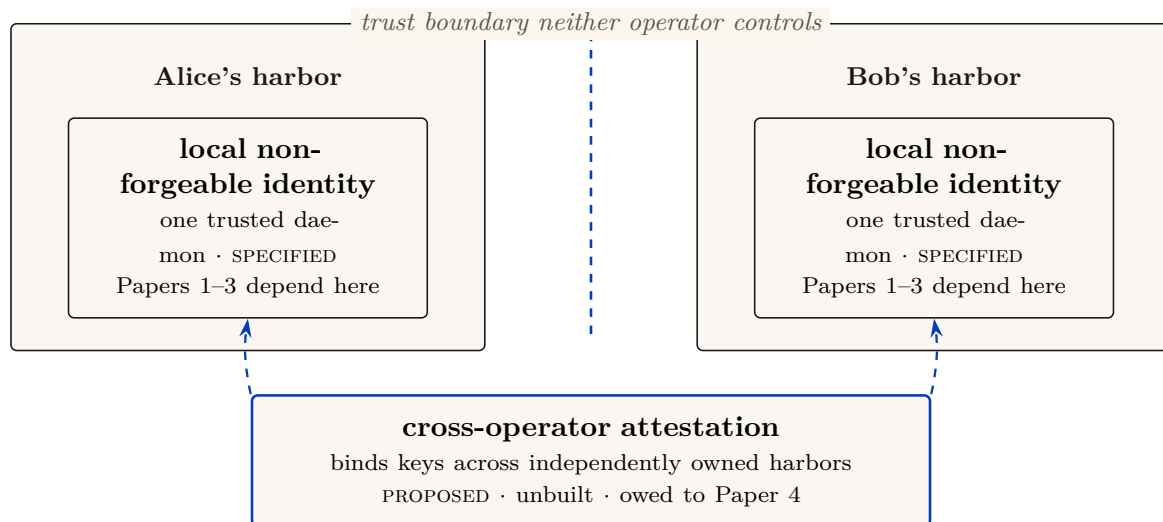


Figure 7: The keystone is two stones. A *local non-forgeable identity* secures identity *within one trusted daemon* — exactly what Papers 1–3 need and assume. The cross-operator boundary needs *cross-operator attestation*, which is unwritten — a federated witness log gives *log-integrity*, not *identity-binding*. This paper depends on the solid stone and hands the market paper the unbuilt one *by name*, rather than borrowing the single-operator threat model across the boundary.

Key idea. The keystone that holds up *this* paper (identity inside one trusted daemon) is *not* the keystone that holds up the cross-operator market. This paper stands on a local non-forgeable identity. The market paper needs cross-operator attestation and must declare it *proposed* rather than borrow the single-operator threat model. This is the clean boundary the volume's cross-layer review demanded.

The second stone's envelope has been standardized — the stone has not. While this volume was being written, the agentic-payments industry shipped the credential dialect the second stone will be carved in: the Agent Payments Protocol [28], adopted by the Universal Commerce Protocol [27] (Google/Shopify, 2026), expresses principal authorization as W3C Verifiable Credentials in SD-JWT form — SD-JWT+kb with hardware-bindable ES256 keys for the principal, JWS detached-content signatures over JCS-canonicalized payloads for the counterparty, keys published as JWK sets under a */.well-known* profile. The reference implementation adopts this envelope at the harbor boundary (ADR-0094), and the honesty rider matters here more than anywhere: *a standard format shrinks the problem; it does not solve it*. What the profile buys is that a principal mandate verifies with off-the-shelf tooling, that principal keys can live in secure enclaves, and that key *distribution* reduces to fetching a cacheable HTTPS document. What it does not buy is the binding of Definition 7.2: a JWK set authenticates keys to an *origin*, not to a principal, and origin-binding across mutually-distrusting operators is exactly the unsolved attestation problem — exercise (3)'s “shared root of trust” in its 2026 industrial form. Cross-operator attestation remains *proposed*; it is now *proposed* with a standardized serialization, which is progress of the unglamorous kind.

Exercises

Check. (1) In one sentence each, state what a local non-forgeable identity provides and what cross-operator attestation would have to add.

Trace. (2) Take any claim in the market paper of the form “Alice’s harbor and Bob’s harbor settle across a boundary neither controls.” Which stone does it secretly assume? What is the actual security guarantee if only the local non-forgeable identity exists?

★ *Open.* (3) Can a federated witness log ever substitute for identity-binding? Argue why log-integrity $\neq$ key-binding, then sketch the minimal additional assumption (a shared root of trust? a sponsor chain? a transparency log) that would close the gap, and the cost each imposes.

8 Reputation as a substrate, not a score

Here is the hinge of the whole volume, stated as the canonical cross-paper sentence:

The reputation estimator is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.

A team tempted to start an agent economy by “building an Elo leaderboard for backends” would be building the cheap, last part first. Elo, Bradley–Terry, and TrueSkill are well-understood; you can implement any of them in an afternoon over a table of game results. The hard, prior, load-bearing work is the three links *beneath* the estimator: a non-forgeable identity (§6), continuity (§5), and an *oracle-witnessed* outcome ledger (Def. 5.1). Without those, the estimator scores noise — or worse, scores a number an adversary controls.

8.1 The estimator family (the well-understood part)

We catalog the estimators not because they are hard but because choosing the *wrong* one for the signal type is a real error.

- **Pairwise / tournament signal** → **Bradley–Terry / Elo / TrueSkill**. When the witnessed signal is “*A beat B on this task*,” the **Bradley–Terry** model [8] is the full-history MLE, and **Elo** [9] is its online special case. A harbor that retains the *whole* history of every settled outcome lives in the Bradley–Terry regime, not the streaming-Elo regime; BT gives more stable ratings than Elo’s recency-weighted update when the population is static. **TrueSkill** [10] adds a calibrated uncertainty σ — the principled way to say “we have not tested this backend much yet,” which matters at cold start.
- **Trust-propagation signal** → **EigenTrust**. When the signal is “who does the network trust,” **EigenTrust** [11] computes a global trust vector as the principal eigenvector of the local-trust matrix — the reputation-systems analogue of PageRank, and the canonical answer to “aggregate peer opinions while resisting collusive cliques.” It is the right tool for the *federated* reputation Paper 4 needs (trust that travels across harbors), and it is *not* a bandit.
- **Marketplace feedback signal** → **feedback systems**. Resnick–Zeckhauser [12] measured a $\sim 8\%$ reputation price premium on eBay; Tadelis [13] frames feedback systems as the platform’s answer to adverse selection. This is the empirical evidence that the third side of the market has real value to price.

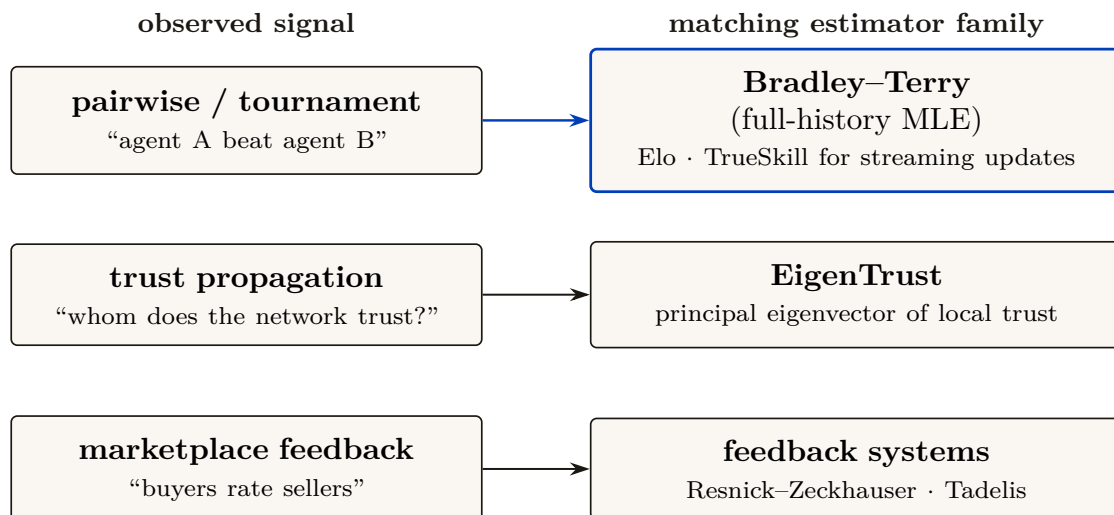


Figure 8: The estimator family, by signal type — the wrong choice is a real error. Pairwise outcomes call for Bradley-Terry / Elo / TrueSkill; network trust calls for EigenTrust (PageRank for reputation — *not* a bandit); marketplace feedback calls for feedback systems with measured price premia (Resnick-Zeckhauser’s ~8%; Tadelis on adverse selection). A SQLite harbor holds the *whole* settled-outcome history, so it is in the Bradley-Terry regime, not the streaming-Elo regime. None of these is the hard part: the estimator is *cheap and last*, while the substrate beneath — non-forgeable identity, continuity, an oracle-witnessed ledger — is the gate.

8.2 Score as telemetry; gate on predicates

Key idea. Publish the score as *telemetry* (a signal the operator and the buyer can read), but *gate* on *predicates* (machine-checkable witnessed facts: “passed the acceptance test,” “no slashed bond in the last k settlements”). The instant the score itself becomes the gate, it becomes a Goodhart target [19]: “when a measure becomes a target it ceases to be a good measure.”

Exercises

Check. (1) Why is a full-history harbor in the Bradley-Terry regime rather than the streaming-Elo regime?

Trace. (2) You have one task with a clear binary acceptance test, and another judged only by a human’s aesthetic preference. Which estimator fits each, and why does EigenTrust fit neither directly?

★ *Open.* (3) The substrate-vs-score sentence says the estimator is “cheap and last.” Construct a scenario where a *better estimator* still buys nothing because the substrate is rotten (e.g. forgeable identity or an agent-authored oracle). Generalize the lesson.

9 Why reputation is *not* a bandit problem

It is tempting — and the routing literature invites the temptation — to model “which agent, model, or skill should I pick?” as a **multi-armed bandit** [15]: arms are agents, pulling an arm yields a reward, and Thompson sampling [16] or UCB balances exploration against exploitation. That framing is correct for one narrow internal task and catastrophically wrong for reputation as a whole. The distinction is load-bearing enough to state as a claim.

Internal routing — *one operator privately choosing which of their own agents to spawn next* — is bandit-shaped. Reputation — *a public, adversarial, multi-dimensional, persistent substrate other parties trade on* — is not. Modeling reputation as a bandit imports four assumptions that are all false in a market.

Bandit assumes...	...but reputation is...	The right tool instead
Stationary arm rewards	Non-stationary: models upgrade, skills version, reputation bubbles build & collapse (Liu-Skrzypacz)	bounded-memory reputation dynamics; version-binding
One scalar reward	Multi-dimensional: accuracy / aesthetics / efficiency, in tension, buyer-weighted	a quality <i>vector</i> + buyer preference (§10)
Non-strategic environment	Strategically adversarial: Goodhart, wash-trading, whitewashing, collusion	mechanism design (Nisan et al.), bonds & slashing
Observed reward signal	Produced by a capturable oracle with its own incentives	bond-and-slash judges; rate-the-raters (§11)

What survives the bandit framing: cold-start *exploration*. TrueSkill's σ and a UCB bonus are the same idea — “test the under-measured arm.” So *one operator's private routing* (choosing which of *their own* agents to spawn next) is legitimately bandit-shaped. The error is exporting that frame to the *public substrate*, where all four breaks bite at once.

Figure 9: Why reputation is not a bandit problem. A multi-armed bandit imports four assumptions — stationarity, a scalar reward, a non-strategic environment, and an observed reward signal — all false for a public reputation substrate. Exploration (one operator's private routing) is the one place the bandit framing legitimately survives.

9.1 The four broken assumptions

1. **Stationarity.** Bandit regret bounds assume the reward distribution of each arm is (near-)stationary. Agent quality is *non-stationary* by construction: models are upgraded, skills are versioned, prompts drift, and a skill's reputation built on v1 says nothing about v2. The Liu-Skrzypacz line of work on *reputation bubbles* [17] shows that with bounded memory and non-stationary types, reputation can build and *collapse* in cycles — behavior a stationary-arm bandit cannot represent.
2. **One scalar reward.** A bandit optimizes a single scalar. Quality is *multi-dimensional*: accuracy, aesthetics, and efficiency are different axes, often in tension (the fast cheap answer is rarely the most beautiful), and a buyer's weighting of them is a *preference vector*, not a constant. Collapse them into one scalar and you have re-created the Goodhart target the substrate argument warned against (§10).
3. **A non-strategic environment.** Bandit arms do not *try* to look good. Agents and their principals are *strategic adversaries*: they will Goodhart the measured axis, wash-trade fake outcomes, whitewash a bad record, and collude. A reputation mechanism is a *game*, and the relevant tool is mechanism design [18], not regret minimization.
4. **A trusted reward signal.** The deepest break: a bandit *observes* its reward. In a market, the reward (the grade) is *produced* by an evaluator who has their own incentives. The grading oracle's incentive-compatibility is not a side-issue — it is a hole in the core theorem (§11). A bandit framing simply assumes the hole away.

9.2 What survives the bandit framing

To be fair to the bandit literature: cold-start *exploration* is real, and the right place to borrow from bandits is the *exploration bonus*. TrueSkill’s σ and a UCB-style bonus are the same idea — “test the under-measured arm.” So *within one operator’s private routing*, a contextual bandit conditioned on the operator’s cost/quality preference vector is a fine engineering choice. The error is exporting that frame to the *public substrate*, where non-stationarity, multi-dimensionality, strategy, and an untrusted reward signal all bite at once.

Pitfall. “We’ll just run Thompson sampling over the agents” is the most common way an agent-economy design quietly assumes its hardest problems away. It assumes a trusted reward (no judge-capture), a scalar reward (no aesthetics-vs-efficiency tension), a non-strategic environment (no Goodhart, no wash-trading), and stationarity (no model upgrades). All four are false in a market.

Exercises

Check. (1) Name the four bandit assumptions and the market reality that breaks each.

Trace. (2) Identify the *one* place a bandit *is* the right tool in this paper, and explain why it survives the four objections there.

★ *Open.* (3) Chiu-Zhang-van der Schaar [21] show competitive agents over-invest in self-improvement, burning surplus despite individual rationality. Does capping the reputation signal’s marginal return restore efficiency, or merely relocate the waste? This is the self-improvement arms-race open problem.

10 The multi-dimensional reputation protocol

We now define the central protocol contribution of this paper: a **multi-dimensional reputation** built over witnessed outcomes. It has three parts: a multi-dimensional quality vector; a market of neutral, conflict-free, influence-free evaluators; and a skill→agent helpfulness attribution. We state it honestly as a design (*SPECIFIED-to-proposed*): the reference system has no reputation component yet, only the witnessed-outcome substrate the protocol scores over.

10.1 Multi-dimensional quality (different axes, different judges)

Definition 10.1 (Quality vector). A witnessed outcome carries a **quality vector** over three axes:

$$\mathbf{q} = (q_{\text{acc}}, q_{\text{aes}}, q_{\text{eff}}).$$

The axes are *accuracy* (did it do the right thing, against an oracle), *aesthetics* (is the artifact good — code clarity, design taste — judged by an expert), and *efficiency* (tokens, wall-clock, dollars, drawn from a cost meter that the reference system already records, **implemented**). Each axis is scored by a *different judge*, because the competent judge of accuracy (a test oracle) is not the competent judge of aesthetics (a senior reviewer) is not the competent judge of efficiency (a meter).

The buyer reads the *vector* and applies their own weighting (the operator’s “cheap vs. careful” preference). Publishing a vector and letting the buyer weight it is the natural disclosure form — and it is precisely what avoids re-collapsing quality into one Goodhart-able scalar. The open design question of *whether* a vector disclosure lets agents Goodhart the most-weighted axis is named as a starred exercise.

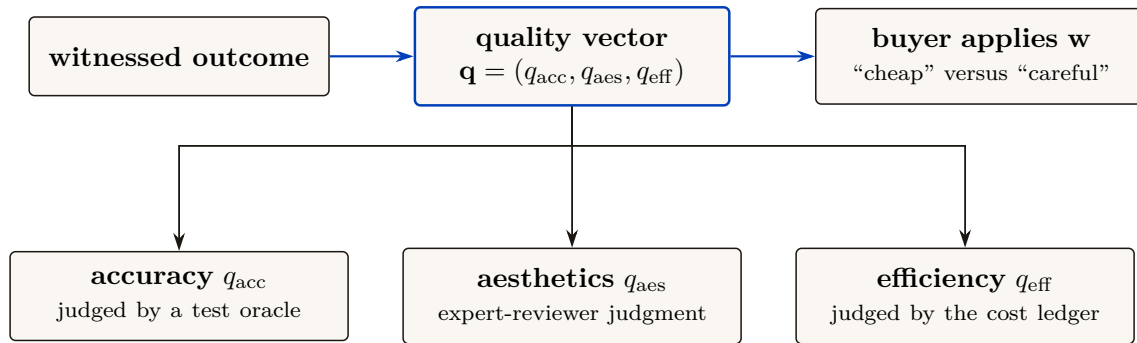


Figure 10: Multi-dimensional reputation. A witnessed outcome carries a quality *vector* $\mathbf{q} = (q_{\text{acc}}, q_{\text{aes}}, q_{\text{eff}})$. Each axis is scored by a *different* competent judge: a test oracle for accuracy, a senior reviewer for aesthetics, the cost meter for efficiency. The buyer reads the vector and applies their own weights — which is exactly what avoids collapsing quality into one Goodhart-able scalar.

10.2 The neutral-judge market (the harbor’s universities and rating agencies)

A judge that grades the work of a party it has a stake in is not a judge. The metaphor the volume uses is exact: a reputation needs the equivalent of *universities* (which certify competence) and *rating agencies* (which rate counterparties) — third parties whose value is precisely that they are *neutral*. We make “neutral” a checklist, not a vibe:

- **Conflict-free.** The judge has no bond, stake, or principal-link in the work being graded. (Excludes self-grading and same-principal grading.)
- **Influence-free.** The judge cannot be paid *by the graded party* for a better grade. Funding is the hard part: a paid judge has a client, an unpaid judge does not scale. The resolution (§11) is that judges post their *own* bond and carry their *own* reputation.
- **De-biased, for LLM judges.** An LLM-as-judge [14] carries position, verbosity, and *self-enhancement* bias (a backend rates its own family up). The discipline is blind, order-swap, pairwise, and family-exclude scoring — never let a backend judge its own family unblinded.

Crucially, an expert third-party judge is *hired through the same bonded ceremony the market uses for any other task*: judging is itself a planned, bonded job, so the harbor that runs the market also runs the hiring of the judges who keep it honest. We state the hiring as a numbered protocol.

Protocol 10.1. The judge-hiring ceremony

To grade a settled outcome o produced by actor a under principal p :

1. **Eligibility filter.** The daemon assembles the candidate judge pool and removes every judge sharing a principal-link, bond, or stake with p (conflict-free), and every judge in a 's model family for the aesthetics axis (de-biasing). *Capability* $\neq$ *permission*: a candidate may be *able* to grade and still be *ineligible*.
2. **Selection.** A judge J is drawn from the eligible pool by a public, conflict-free policy (e.g. reputation-weighted sampling on the judging axis), so neither a nor p can choose their own grader.
3. **Bond post.** J posts a bond B_J before seeing the work. The bond is recorded in the outcome ledger and locked for the dispute window.
4. **Blind grading.** J scores the relevant axis under the de-biasing discipline (blind, order-swapped, pairwise where possible). The grade enters the ledger as a witnessed event keyed on J 's non-forgeable identity.
5. **Settlement and exposure.** The outcome settles on the grade; the bond flow follows. J 's bond remains exposed to a sampled **re-audit** (§11): a fresh, conflict-free judge who overturns J 's grade *slashes* B_J and updates J 's own reputation downward.

Figure 11 draws this: graded work flows to a judge selected for conflict-/influence-freedom, the judge posts a bond, the grade enters the outcome ledger, and a later re-audit can slash a judge whose grade is overturned.

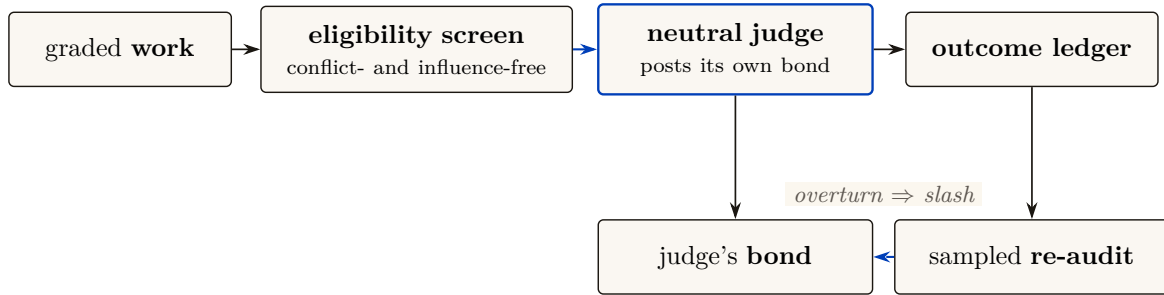


Figure 11: The neutral-judge market. Graded work flows to a judge selected for conflict- and influence-freedom; the judge posts its own bond, the grade enters the outcome ledger, and a sampled re-audit can slash a judge whose grade is overturned. “Neutral” is a checklist (conflict-free, influence-free, de-biased), not a vibe, and the judge is hired through the same bonded ceremony the market uses for any task (Protocol 10.1) — the judge’s own reputation is the asset it protects.

10.3 S3 — a graded job, end to end

Make Protocol 10.1 concrete. Bob has rented Alice’s agent *Heron* to refactor an authentication module against a fixed acceptance suite, under a posted bond. Heron delivers; the outcome o enters Bob’s harbor as a *settled but ungraded* delivery, and the daemon runs the ceremony. *Three* judges are hired, one per axis, because the axes need different competence:

- **Accuracy** is graded by a *test oracle* — the acceptance suite Heron cannot author — which returns $q_{acc} = 1.0$: every test passes.
- **Aesthetics** is graded by a hired senior-reviewer judge, *Mara* (drawn from the eligible pool, in neither Alice’s nor Bob’s principal, not in Heron’s model family). Mara posts her bond, reads the diff blind, and returns $q_{aes} = 0.7$: correct but over-clever, with a fragile regex she flags.
- **Efficiency** is graded by the cost meter: $q_{eff} = 0.9$, cheap and fast.

The quality vector $\mathbf{q} = (1.0, 0.7, 0.9)$ lands in Heron’s outcome ledger. Bob, who weights aesthetics heavily for security-critical code, reads the *vector* (not a collapsed scalar), sees the 0.7, and decides to keep Heron but require a second reviewer on auth work — a decision the scalar would have hidden. Two weeks later a routine re-audit re-grades a sample of Mara’s recent jobs and finds she rubber-stamped a different agent’s insecure change as $q_{aes} = 0.9$. The re-auditor overturns that grade; Mara’s bond on *that* job is slashed and her judging reputation drops — exactly the exposure step of Protocol 10.1. Note what made every step possible: each grade is keyed on a non-forgable identity, so neither Heron nor Mara can shed the record by respawning.

10.4 Skill → agent helpfulness attribution

A distinctive, tractable, and *shippable-soon* signal: track whether a **skill** — a reusable unit of agent know-how (a structured instruction document, its referenced material, and any scripts it carries) — actually *helped* an agent, *especially when that skill’s content was loaded into the agent’s context window*. The witnessed event is concrete: skill s was grafted into agent a ’s context at task t ; t later settled with quality vector $\mathbf{q}$. Over many such events we estimate a *skill-helpfulness* effect — did loading s raise $\mathbf{q}$ relative to comparable tasks without it? This is reputation for *tools*, keyed on witnessed, context-attributable outcomes, and it is the licensing side’s substrate that the market paper consumes. It is also the cleanest near-term instance of the whole thesis: a non-forgable event (the graft), a witnessed outcome (the settled task), and a multi-dimensional grade — no bandit required. We pin the event down as a schema.

Protocol 10.2. The skill-helpfulness witnessed-event schema

Each graft of a skill into an agent’s context emits an append-only event with fields:

- **skill_id**, **skill_version** — the tool and its exact version (a portfolio for v1 says nothing about v2).
- **agent_id** — the *non-forgable* identity that received the graft, so the event cannot be re-attributed by respawn.
- **task_id**, **task_descriptor** — a content hash and a feature vector of the task, used to assemble a comparison set of similar tasks *without* s .
- **grafted_at**, **context_window_ref** — evidence that s ’s content actually entered the context (not merely that s was available).
- **outcome_ref**, **quality_vector** — the witnessed settlement and its multi-dimensional grade, written only after the task closes against an oracle.

A buyer estimates helpfulness as the difference in $\mathbf{q}$ between matched tasks with and without s in context — verifiable from the events alone, without trusting the skill’s author.

Pitfall. Attribution is not causation. “Skill s was in context and the task succeeded” does not prove s caused the success. The honest estimate needs a comparison set (similar tasks without s , matched on **task_descriptor**) and is still observational. Sell it as *helpfulness evidence*, not *proof of value*, and gate licensing clawbacks on the witnessed green/red settlement, not on the helpfulness estimate alone.

Exercises

Check. (1) For each of accuracy / aesthetics / efficiency, name a competent judge and an *incompetent* one.

Trace. (2) Walk a judging task through the judge-hiring ceremony (Protocol 10.1): who plans it, who posts the bond, where does the grade land, and what slashes the judge?

Trace. (3) Design the witnessed event for skill-helpfulness. What fields does it need so that a buyer can verify “this skill helped” without trusting the skill’s author?

★ *Open.* (5) If you publish the 3-vector and buyers weight it, do agents Goodhart the most-weighted axis? If you publish a scalar, you have recreated the bandit framing the substrate argument rejects. What is the right disclosure form for vector reputation?

11 The grading oracle’s incentive-compatibility

Every folk-theorem incentive-compatibility result in the L3 economy bottoms out in “the daemon derives the grade” — but the grade is produced by a *capturable evaluator*. This is not a side-gap. It is a hole in the *core theorem*, and we treat it head-on rather than assuming it away.

Property 11.1 (The grading-oracle hole). Let an outcome’s settlement (and thus the bond flow and the reputation update) depend on a grade g produced by an evaluator J . If J can be captured — bribed, colluded with, or self-interested — then the incentive-compatibility of *every* mechanism that settles on g is only as strong as J ’s own honesty. Assuming J honest is assuming the conclusion.

There are two honest ways to close the hole, and a recursion to bound.

Option A — bond-and-slash the judges. Make judging a bonded role (§10): J posts a bond before grading; a later **re-audit** (a fresh, conflict-free judge, sampled) that overturns J ’s grade *slashes* J . Now lying has an expected cost, and a judge’s own reputation is the asset they protect. This converts “trust the oracle” into “the oracle has skin in the game,” which is the same move slashing makes for agents.

Option B — 2-of-3 multi-oracle. Settle disputes against a quorum of independent oracles (automated test / evidence review / human), so capturing the grade requires capturing a majority. This raises the cost of capture but does not eliminate it, and the *human* oracle is a scarce, expensive, capturable resource in its own right (the economics of arbitration capacity is a named open problem, below).

The rate-the-raters recursion — does it terminate? Option A pushes the problem up a level: who audits the auditors? The re-auditor is itself a judge with incentives. The recursion “rate the raters who rate the raters” must be shown to *terminate* or to *contract*. We do not claim a proof; we claim the shape of one and mark the rest open.

CONJECTURED CONTRACTION TOWARD AN HONEST FIXED POINT

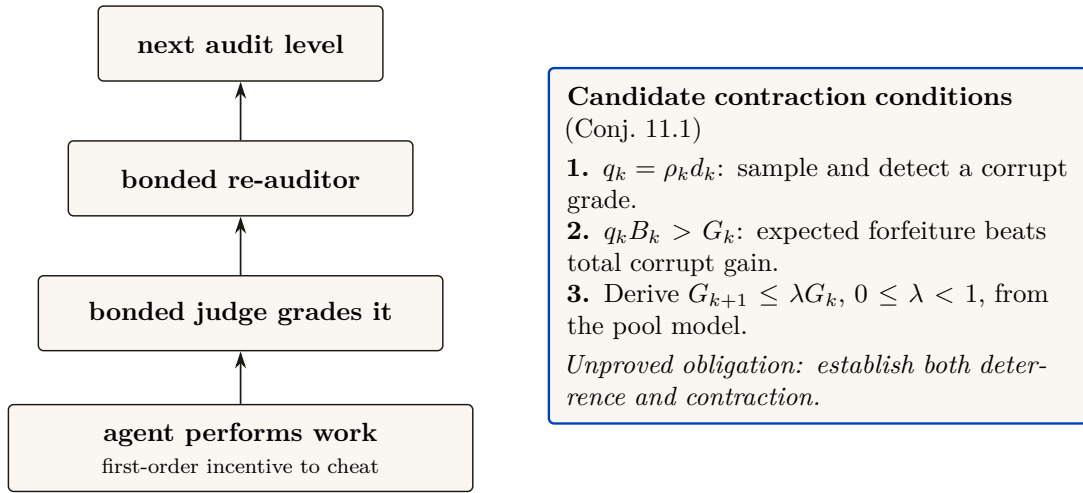


Figure 12: The rate-the-raters recursion. Conjecture 11.1 separates two obligations that a recursive audit design must prove: expected forfeiture $\rho_k d_k B_k$ must exceed the judge’s total corrupt gain G_k at each level, and the pool model must make the remaining corruption opportunity shrink by a uniform factor $\lambda < 1$. A bond larger than one bribe is not enough when re-audit is probabilistic. Neither obligation is depicted as established.

Conjecture 11.1 (Contractive re-audit conditions). At audit level k , let G_k bound the judge’s total one-shot gain from a dishonest grade (bribes plus private side benefits), let ρ_k be the probability of re-audit, let d_k be the probability that a re-audit detects and overturns the dishonest grade, and let B_k be the slashable bond. Local deterrence requires the expected-forfeiture inequality

$$\rho_k d_k B_k > G_k.$$

Contractive re-audit additionally requires a pool-and-sampling argument that derives a uniform $\lambda < 1$ with $G_{k+1} \leq \lambda G_k$, or a finite stopping rule with the same incentive effect. A conflict-free pool larger than one clique is a candidate ingredient, not by itself such a proof. Establishing both the inequality and the contraction bound at every reachable level is the open obligation.

Pitfall. Conjecture 11.1 states required conditions, not a proof that any deployed pool satisfies them. It is the precise statement of what the market paper’s central incentive-compatibility result *needs* and currently *lacks*. The contribution here is to name the hole exactly, give two candidate closures, and state the termination obligation — *not* to claim it is closed. Honest labels apply: this is *proposed*.

Key idea. The grading oracle is the load-bearing assumption hidden under every IC claim in the economy. “The daemon derives the grade” only helps if the grade’s *producer* is itself incentive-compatible. Either bond-and-slash the judges and prove the rate-the-raters recursion contracts, or accept that the core theorem is conditional on an honest oracle — and say which.

Exercises

Check. (1) State Property 11.1 and explain why “the daemon derives the grade” does not, by itself, close it.

Trace. (2) Walk Option A: a judge grades dishonestly, a re-audit overturns it. Follow the bond flow and the reputation update for the judge.

★ *Open.* (4) Prove or refute Conjecture 11.1. If the re-audit probability ρ is too low, the tower is not contractive — find the critical ρ^* as a function of bond size and bribe ceiling.

★ *Open.* (10) Arbitration capacity: human oracles are scarce. Design a market for bonded, rated, slashable arbiters and argue whether it converges to honest rulings or to rubber-stamping under load.

12 Revocation, tombstones, and bounded memory

A reputation built on witnessed outcomes must answer two questions the cheerful “reputation never ends” framing dodges: what happens when a witnessed outcome turns out to be *fraudulent*, and is no-decay actually *good*? We adopt the volume’s reconciled position on both, against the naive version.

12.1 Monotone in honest outcomes, individually revocable

Property 12.1 (Reputation monotonicity with tombstones). Reputation is **monotone in honest witnessed outcomes** — a genuine delivery is never silently erased by the passage of time — *but* a witnessed outcome is itself **individually revocable via a propagating tombstone**: a compensating, append-only entry that retracts a specific outcome later found fraudulent (a colluding judge, a faked oracle). “Never decays” applies to honest outcomes *only*; it is *not* an anti-revocation property.

This is a **saga-style** compensation [20]: you do not mutate the ledger in place (that would break append-only and auditability); you append a tombstone that nullifies the targeted entry, and the tombstone *propagates* across harbors bounded by the federation’s revocation-convergence window (the market paper’s job). The poisoned data point was spendable only during the propagation window; bounding that window is the federation’s responsibility. We state the retraction as a protocol.

Protocol 12.1. Tombstone revocation

To retract a witnessed outcome o later found fraudulent:

1. **Trigger.** A re-audit (Protocol 10.1, step 5) or a dispute overturns the grade on o , establishing fraud against an oracle.
2. **Append, never mutate.** A *tombstone* entry $\bar{o}$ is appended to the ledger, carrying o ’s id, the overturning evidence, and the identity of the retracting judge. The original o stays in place; auditability is preserved because nothing is erased.
3. **Recompute.** Every reputation score that consumed o is recomputed with o nullified by $\bar{o}$. Because the estimator is cheap (§8), recomputation is not the bottleneck.
4. **Propagate.** $\bar{o}$ gossips to every harbor that imported o , bounded by the federation’s revocation-convergence window τ . Each harbor applies step 3 on receipt.
5. **Bound the damage.** The fraudulent reputation was spendable only during τ ; the protocol’s job is to make τ short and provable, not to pretend the window is zero.

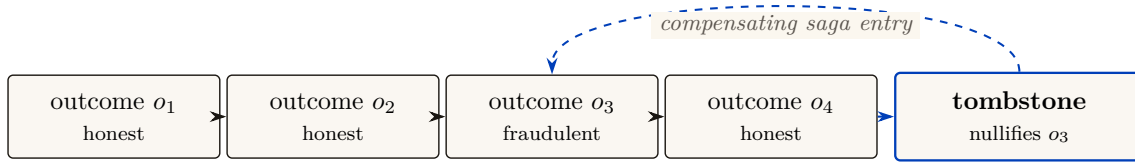


Figure 13: Tombstone revocation on an append-only ledger (never mutate in place). A fraudulent outcome is not deleted (that would break append-only and auditability); it is nullified by an appended *tombstone* — a saga-style compensating entry — which propagates across harbors bounded by the revocation-convergence window (Paper 4); the poisoned point was spendable only during that window. “Reputation never ends” is corrected to: monotone in *honest* outcomes (they are never silently erased), individually revocable via tombstone, with bounded memory (Liu–Skrzypacz) as a tunable parameter rather than a virtue — ∞ -memory freezes types and removes the incentive to keep performing.

12.2 S4 — the fraudulent outcome retracted

Recall S3: the re-audit caught Mara rubber-stamping a different agent’s insecure change as $q_{\text{aes}} = 0.9$. Follow that one outcome through Protocol 12.1. The graded agent — call it *Crane*, running under a third operator — had colluded with Mara: Crane shipped a known-insecure diff and Mara graded it high so Crane’s aesthetics reputation would clear Bob’s hiring bar. For two days it worked: Crane’s vector showed (0.95, 0.9, 0.8) and Bob nearly hired it. Then the sampled re-audit re-graded the diff blind, found the injection, and overturned Mara’s grade. A tombstone $\bar{o}$ is *appended* (the original stays, for the audit trail); every score that consumed Crane’s inflated q_{aes} is recomputed with it nullified; Mara’s bond on that job is slashed and her judging reputation drops; Crane’s collusion is itself recorded as a fraudulent outcome against *its* non-forgable identity, which it cannot shed. The tombstone gossips to Bob’s harbor within the convergence window, so by the time Bob next reads Crane’s vector the 0.9 is gone. The fraud was real and briefly spendable — the honest claim is that the window was *bounded*, not that it never existed.

12.3 Bounded memory is a design parameter, not a virtue to assert away

The naive “no decay window an agent can outrun” framing sounds principled but is mechanism-design-hostile. Liu–Skrzypacz [17] show that with *bounded* memory, reputation can be *welfare-improving* relative to infinite memory — and that infinite memory (∞) is rarely optimal, because it freezes types and removes the incentive to keep performing. Bounded memory is a **tunable parameter**, and the right setting is an empirical/mechanism-design question, not a slogan.

Pitfall. “Reputation never ends” as a *virtue* is wrong twice over: it makes a fraudulent outcome permanent (needs the tombstone of Property 12.1), and it asserts ∞ -memory is optimal when the reputation-bubble literature shows it usually is not. Engage bounded memory as a *parameter*; do not assert no-decay as a feature.

Key idea. Two corrections to the cheerful framing: (1) reputation is monotone in *honest* outcomes but *individually revocable* via a propagating tombstone — a saga compensation, not in-place mutation; (2) bounded memory is a *design parameter* (Liu–Skrzypacz reputation bubbles), not an anti-feature to assert away.

Exercises

Check. (1) Why must revocation be a tombstone (append) rather than a delete (mutate) on the outcome ledger?

Trace. (2) A judge is later found to have colluded on five settled outcomes. Walk the tombstone for one of them: what is appended, what propagates, and what bounds the window the poisoned reputation was spendable?

★ *Open.* (8) Reputation-claim revocation with an append-only, bounded-memory ledger across N harbors: how do you prove the correction converged and bound the spendable window, without violating append-only? This is the cross-harbor analogue of capability revocation (Paper 4).

13 What this paper hands the market (Paper 4)

This paper is the bridge; its output is a precise hand-off to *The Harbor Economy* (Paper 4). We summarize what is delivered, what is borrowed, and what is explicitly owed.

Artifact	Maturity	Role in the market paper
Role / person distinction (Defs. 3.1–3.2)	—	The third side of the market is a <i>tradeable person</i> , defined here.
Principal-as-economic-entity (Def. 6.2)	SPECIFIED	Bonds and sanctions key on the principal; the market paper cross-references, not redefines.
The three continuity organs	implemented PARTIAL SPECIFIED	Reputation keys on organ 3 (the outcome ledger).
Outcome ledger / oracle (Def. 5.1)	SPECIFIED	The witnessed-outcome substrate every settlement updates.
Multi-dimensional reputation (Def. 10.1, Protos. 10.1, 10.2)	SPECIFIED to <i>proposed</i>	The multi-dimensional score the market reads; the licensing side’s signal.
Neutral-judge market	SPECIFIED to <i>proposed</i>	The “universities and rating agencies” that keep grades honest.
Grading-oracle hole (Prop. 11.1, Conj. 11.1)	<i>proposed</i>	<i>Owed back to the market paper</i> : its central IC theorem is conditional on closing this.
Cross-operator attestation (Def. 7.2)	<i>proposed</i>	<i>Named dependency the market paper must build</i> ; not assumed here.
Tombstone revocation (Prop. 12.1)	SPECIFIED	Reputation is revocable; convergence is the market paper’s federation job.

Table 4: The bridge’s hand-off to the market paper. Two items flow *back* as named, unsolved obligations: the grading-oracle’s incentive-compatibility (Conj. 11.1) and cross-operator attestation (Def. 7.2). The volume is one book precisely because these debts are named in the same words on both sides of the seam.

Build the harbor first. The trade comes when the ships do — but the thing that makes the trade possible (continuity that turns spawns into persons) is the same thing that makes the single-player tool good. The bridge is not a detour; it is the load path.

14 Open problems (the starred exercises, collected)

The genuinely unsolved problems this paper surfaces, gathered for the reader who wants the research frontier rather than the pedagogy:

1. **Branching continuity (§4).** If a checkpoint is forked into two live incarnations, which inherits the reputation? Parfit says identity-is-not-what-matters; the economy needs a rule, and the rule invites an attack.
2. **Agent-death taxonomy (§5).** What is recoverable vs. fundamentally lost when an LLM agent dies mid-thought? The hard half of checkpoint-with-teeth.
3. **Cross-operator attestation (§7).** Binding identity keys across harbors you do not own. The highest-leverage unbuilt keystone for the market.
4. **The grading-oracle / rate-the-raters recursion (§11, Conj. 11.1).** Prove the re-audit tower contracts, or accept the core IC theorem is conditional.
5. **Vector-reputation disclosure (§10).** Publish a vector and agents Goodhart the heaviest axis; publish a scalar and you have a bandit. The right disclosure form is open.
6. **Self-improvement arms race (§9).** Chiu et al. 2025: does capping reputation's marginal return restore efficiency or relocate the waste?
7. **Reputation-claim revocation convergence (§12).** Surgically retract a fraudulent outcome across N harbors, prove convergence, bound the spendable window — without violating append-only.
8. **Bond-farming defense (§6).** Principal binding vs. listing fee vs. sampled re-audit of high-velocity pairs; the false-positive cost to honest high-volume counterparties.
9. **Arbitration capacity (§11).** A market for bonded, rated, slashable human/automated arbiters: honest rulings, or rubber-stamping under load?
10. **Skill-version reputation (§10).** A portfolio proof for skill v1 says nothing about v2; the lemons problem reappears at every version bump.

15 Conclusion

We set out to bridge the wedge to the market by answering one question: what has to be true before a coding agent's work can be *sold*? The answer is not a score. It is a *self* — a role plus continuity, keyed on an identity that cannot be re-picked, accumulating a transitive, witnessed history that survives every respawn. The estimator that turns that history into a number is the cheap, last part; the substrate beneath it is the gate, and most of that substrate is honestly not built yet.

We were disciplined about the keystone (this paper stands on a local non-forgeable identity and hands the market paper the unbuilt cross-operator attestation by name), about the hardest hole (the grading oracle's incentive-compatibility is a flaw in the core theorem, not a footnote), and about the cheerful slogans (reputation is monotone in *honest* outcomes but individually revocable, and bounded memory is a parameter, not a virtue). Reputation is not a bandit problem — it is non-stationary, multi-dimensional, strategically adversarial, and graded by an oracle with its own incentives.

A spawn does work and disappears. A person does work and stays — and only what stays can be priced. The harbor’s whole economy is the machinery for manufacturing, and then honestly grading, a self.

A Implementation status of the reference system

The body of this paper is written to stand on its own: every mechanism is described as a component (a memory record, a checkpoint organ, a witnessed-outcome ledger) and graded on the neutral maturity scale (**implemented** / PARTIAL / SPECIFIED / *proposed*). This appendix grounds that scale in a concrete *reference system* — a running coordination daemon for local agent swarms — so a reader who wants to know “which of these actually runs today” can check. Nothing in the body depends on this appendix; it exists so the maturity labels are falsifiable rather than rhetorical.

Component (paper concept)	Maturity	Status in the reference system
Episodic memory (Organ 1)	implemented	Durable typed episodes promoted from execution history; runs in production.
Restart organ (Organ 2)	PARTIAL	Heartbeat-staleness detector that salvages dead agents; forwards a <i>summary</i> , not execution state.
Outcome ledger (Organ 3)	SPECIFIED	Durable-commitment record type exists; oracle-bound closure is specified, not shipped.
Cost meter (efficiency axis)	implemented	Per-task token/wall-clock/dollar accounting recorded today.
Local non-forgable identity	SPECIFIED	Daemon-minted opaque id bound to a credential; specified. Identities are self-asserted strings today.
Capability jail (capability $\neq$ permission)	SPECIFIED	Per-agent tool-allowlist and scoped filesystem; specified.
Honest-attestation discipline	implemented	“Report all-good only when verified” is an enforced engineering rule.
Multi-dimensional reputation	SPECIFIED to <i>proposed</i>	No reputation component yet; the substrate it scores over is partly built.
Neutral-judge ceremony (Proto. 10.1)	SPECIFIED to <i>proposed</i>	Bonding and task-planning primitives exist; the hiring ceremony is designed.
Skill-helpfulness schema (Proto. 10.2)	SPECIFIED	Skill grafting is observable; the witnessed-event schema is specified.
Tombstone revocation (Proto. 12.1)	SPECIFIED	Append-only ledger semantics are designed; cross-harbor propagation is <i>proposed</i> .
Cross-operator attestation	<i>proposed</i>	The unbuilt keystone for the market; owed to the companion market paper.

Table 5: Maturity of each paper concept against a reference coordination daemon. The honest summary: the *organs* of continuity exist (one weakly); the *spine* of reputation — a witnessed-outcome ledger keyed on a non-forgable identity — is specified but not yet shipped.

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